Amoeba Sisters | Video Recap

NAME: _____

Amoeba Sisters Video Recap: *Mutations (Updated)*

1. What is a **mutation**?

_____

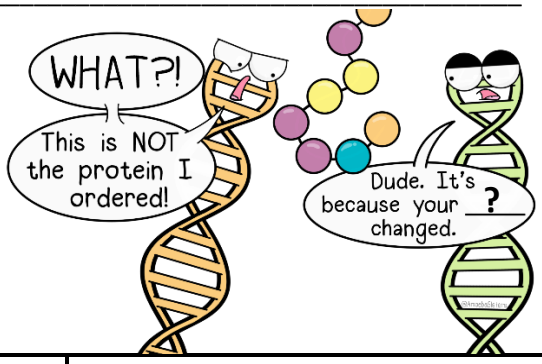
_____

_____

2. A specific part of a **nucleic acid** (such as DNA or RNA) experiences a mutation that could lead to a different protein produced. View the illustration below of DNA. Which part of the DNA experiences the mutation?

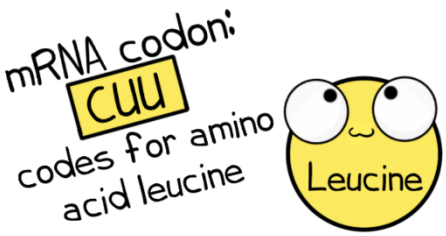
_____

3. On the DNA illustration, draw an *arrow* to show *where* the answer to #2 could be located.



Mutations can be harmful, helpful, or neutral in their effect. A **silent mutation** tends to have a neutral effect as it does not result in coding for a different amino acid. Using your mRNA codon chart, give another mRNA codon that this CUU could mutate to and *still* code for leucine.

4. The mRNA codon CUU could mutate to **C**_____ and *still* code for leucine, which would not change the amino acid.



5. Which type(s) of organism(s) can experience a mutation? Mark any that apply.

_____ Animals (this includes humans)

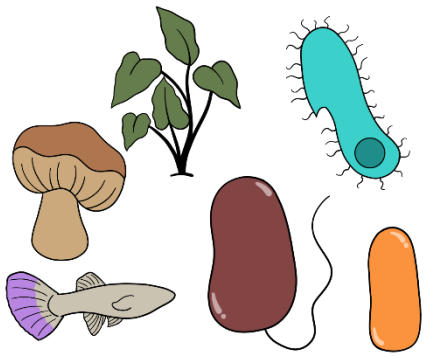
_____ Archaea

_____ Bacteria

_____ Fungi

_____ Plants

_____ Protists



6. Even a **gene mutation** that is a **point mutation**, meaning it affects one nucleotide base, can still make a major change

Consider the below information for normal hemoglobin:

Portion of Hemoglobin DNA	GGA CTC CTC
mRNA	CCU GAG GAG
Amino Acids	Proline-Glutamic Acid-Glutamic Acid

Sickle Cell Anemia is caused by a point mutation known as a **substitution.** Show what would occur *if* the *first* T ("thymine") DNA base in the portion shown above experienced a mutation with a substitution of A ("adenine").

Sickle Cell Hemoglobin:

Portion of mutated hemoglobin DNA: _____

mRNA: _____

Amino Acids:_____

Amoeba Sisters LLC
©All rights reserved

Amoeba Sisters | Video Recap

NAME: _____

Amoeba Sisters Video Recap: Mutations (Updated)

7. An **insertion or deletion** can result in a **frameshift mutation**. To demonstrate this, complete the following. *Note: You will need a codon chart.*

Normal Strand:

DNA: GCA ATG CAC

mRNA: _____

Amino Acids: _____

Deletion (causing a frameshift):

Taking out the first "G" in the original DNA above results in:

DNA: CAA TGC AC

mRNA:_____

Amino Acids:_____

How did the frameshift change the amino acids?

_____

8. Check your understanding! Mark any that are *correct*.

_____Mutations are random.

_____Mutations are mostly beneficial and useful for an organism.

_____Mutations can occur in both DNA and RNA, which are **nucleic acids**.

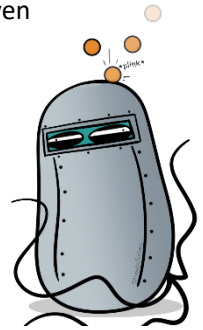
_____Mutations can only occur during **interphase**.

_____Not all genes code for proteins.

_____Not all genes are "turned on" at a given time.

_____**Substitution** mutations typically result in a **frameshift mutation**.

_____Mutations can be **genetically inherited**.



Chromosome Mutations

Sketch It!

Create illustrations to show the following chromosome mutations. Note: Chromosomes exist in both prokaryotic and eukaryotic cells, but prokaryotic chromosome structure tends to be very different from eukaryotic chromosomes.

9. Duplication

10. Deletion

11. Inversion

12. Translocation

Amoeba Sisters LLC
©All rights reserved